

Name: _____
Class: _____
Date: _____

Mr. Rawson • GCSE Physics

RP 16: I-V Characteristics

Key concept

Two weeks ago you measured a nichrome wire and got a **straight line**. Resistance was the same whatever the current — that is what a straight line through the origin *means*. As always,

$$R = \frac{V}{I} \quad \text{resistance } (\Omega) = \frac{\text{potential difference (V)}}{\text{current (A)}}$$

Today's question is whether every component behaves like that wire. It does not. Some components change their resistance as you push more current through them, and the graph is how you see it happen.

Reading an I-V graph (this is the whole skill)

You are plotting **current** up the page against **potential difference** across the page. The gradient is not the resistance — but it tells you about it:

- **Straight line** → resistance is **constant**. Doubling V doubles I .
- **Curve bending towards the V axis** (getting flatter) → current is no longer keeping up with pd, so the resistance is **increasing**.
- **Flat along the axis** → almost no current for a large pd, so the resistance is **very high**.

Why we go negative. You will reverse the supply and take readings with the pd the other way round, so your graph runs through all four quadrants. For two of today's components that half looks exactly like the first half. For the third it does not, and that is the entire point of it.

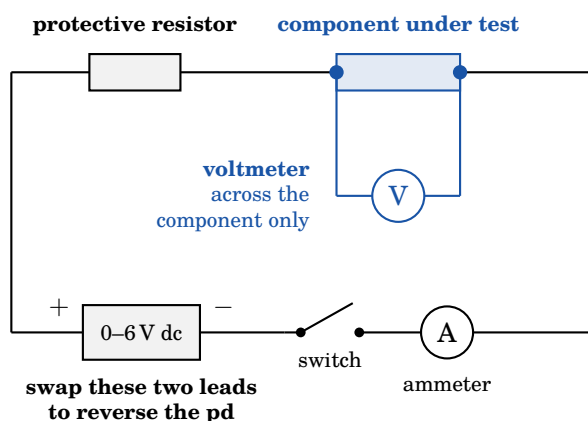
The question

How does the current through a component depend on the potential difference across it — and is the answer the same for every component?

You will test **three**: a fixed resistor, a filament lamp, and a diode.

Independent variable what you change	<i>The potential difference across the component — including reversing it.</i>
Dependent variable what you measure	<i>The current through the component.</i>
Control variables what stays the same	<i>The same component throughout each set of readings; the same protective resistor; the temperature of the fixed resistor — keep the switch open between readings so it does not warm up.</i>

The circuit



One circuit, three components. You build it once and swap only the blue box. The ammeter is in the loop, so the whole current goes through it. The voltmeter goes across the component alone.

The protective resistor is not optional

It is in series with everything, and it is there to **limit the current**.

A diode with nothing to limit it will take a very large current the moment it starts conducting, get hot, and be destroyed — in about a second, and without warning. The fixed resistor in the top-left corner of the diagram is what stops that happening. **It stays in the circuit for all three components**, so every set of readings is treated the same way.

Because it is in series, some of the supply pd is dropped across it. That is fine. Your voltmeter is across the *component*, so what you record is the component's pd and nothing else.

Method

1. Build the circuit above with the **fixed resistor** as the component under test. Leave the switch **open** until you have been checked.
2. Set the supply to its **lowest** setting. Close the switch, read the **voltmeter** and the **ammeter** together, then **open the switch again**.
3. Work up through the supply settings, taking a pair of readings at each. Six or seven pairs is plenty. **Open the switch between every reading**.
4. **Now reverse**. Switch off, swap the two leads at the supply, and repeat the whole set. Record these readings as **negative** — both the pd and the current.
5. **Plot as you go**, straight onto the grids on the next pages. Do not wait until the end. The shape is the result, and you want to see it appearing.
6. Swap the component for the **filament lamp** and repeat steps 2–5.
7. Swap the component for the **diode** and repeat steps 2–5. Read the box on the diode page **before** you start it.

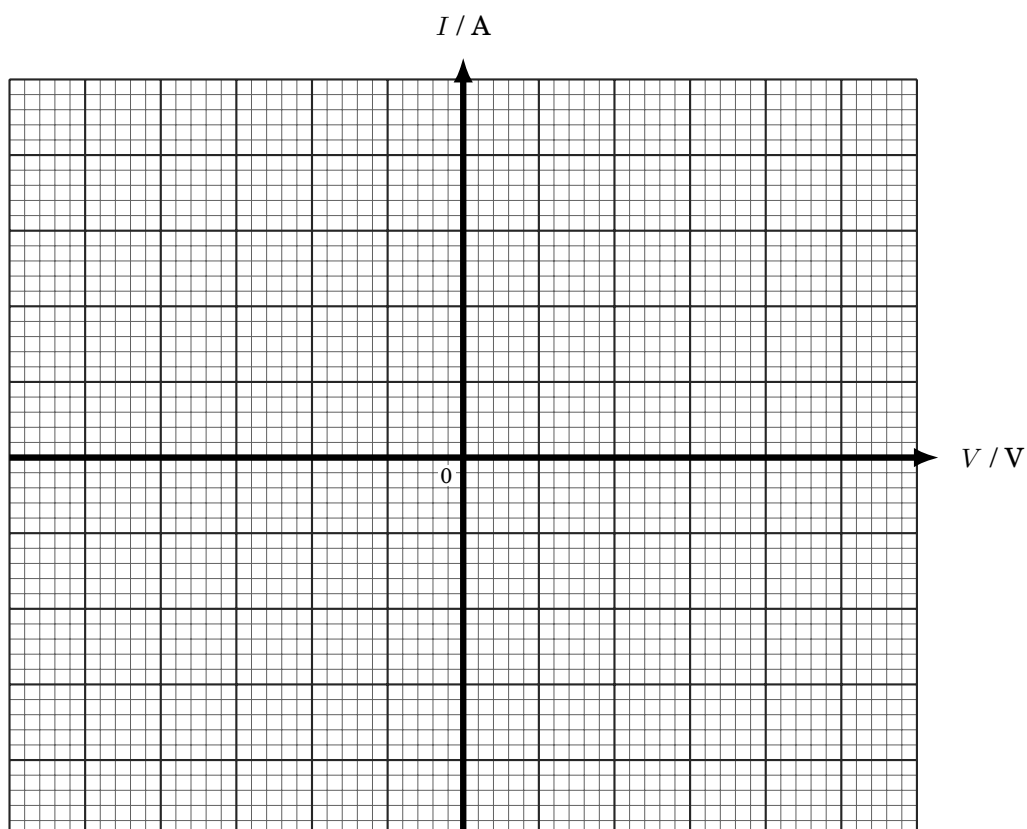
Safety and good practice

- **The switch stays open between readings.** This is the whole reason the fixed resistor counts as being “at constant temperature”. A circuit left closed warms up, and a warm resistor is no longer the component you set out to measure.
- **Never remove the protective resistor.** Not for one reading, not to “see what happens”.

- The filament lamp gets hot — that is the physics, not a fault. Do not touch the bulb.
- Switch off at the supply *before* swapping any component or reversing any lead.
- If anything smells hot or a meter slams to full scale, open the switch and put your hand up.

Component 1 — the fixed resistor

pd V / V	Current I / A	pd V / V	Current I / A
forward		reversed (negative)	



Label your own scale on both axes first — negative to the left of zero and below it. Then plot every point, forward and reversed, and draw **one** straight line through them all.

1) **Describe** the shape of your graph.

A straight line through the origin, running from bottom-left to top-right. Reversing the pd simply continues the same line into the negative quadrant.

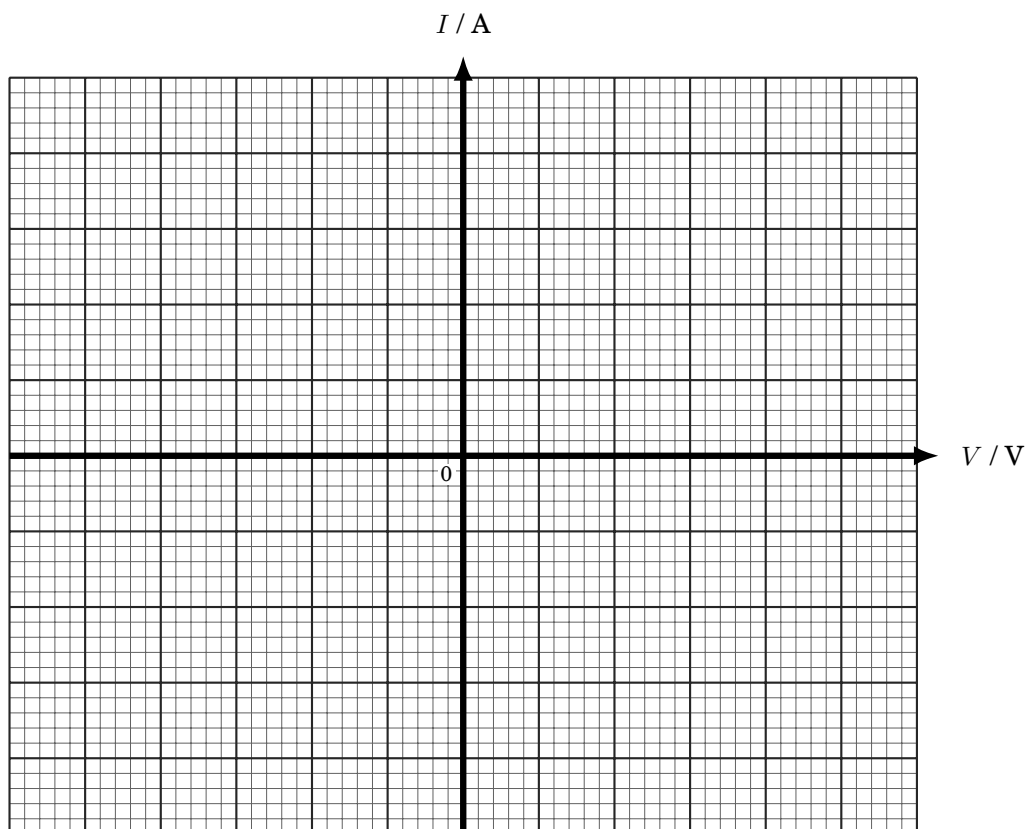
2) **State** what that shape tells you about the resistance of the fixed resistor.

The resistance is constant. Current is directly proportional to potential difference, so V / I gives the same value at every point on the line.

Component 2 — the filament lamp

pd V / V	Current I / A	pd V / V	Current I / A
forward		reversed (negative)	

Use the whole range of the supply for this one. The bend is what you are looking for, and it only shows up once the filament is properly hot. If you stop at 2 V your graph will look straight and you will have measured nothing. Take it as high as the supply allows — the lamp is *meant* to glow.



Label your own scale on both axes first. Then plot every point and draw a **smooth curve** — not a straight line, and not dot-to-dot.

3) **Describe** the shape of your graph, and say how it differs from the resistor's.

An S-shaped curve through the origin. Near the origin it is steep and almost straight, but as the pd increases in either direction the curve flattens, bending towards the horizontal axis. The resistor gave a straight line; this one does not.

4) **Explain** why the curve gets flatter as the potential difference increases. Your answer must mention temperature.

As the pd increases the current increases, and the current heats the filament. The hotter the filament, the more the metal ions vibrate, so electrons collide with them more often and the resistance increases. A larger resistance means the current rises by less for each extra volt — so the graph bends over.

5) **Explain** why the curve has the same shape on both sides of the origin.

The filament heats up whichever way the current flows — the heating does not depend on direction. So the resistance rises in the same way when the pd is reversed, giving a curve with the same shape rotated through the origin.

Component 3 — the diode

Read this before you connect the diode

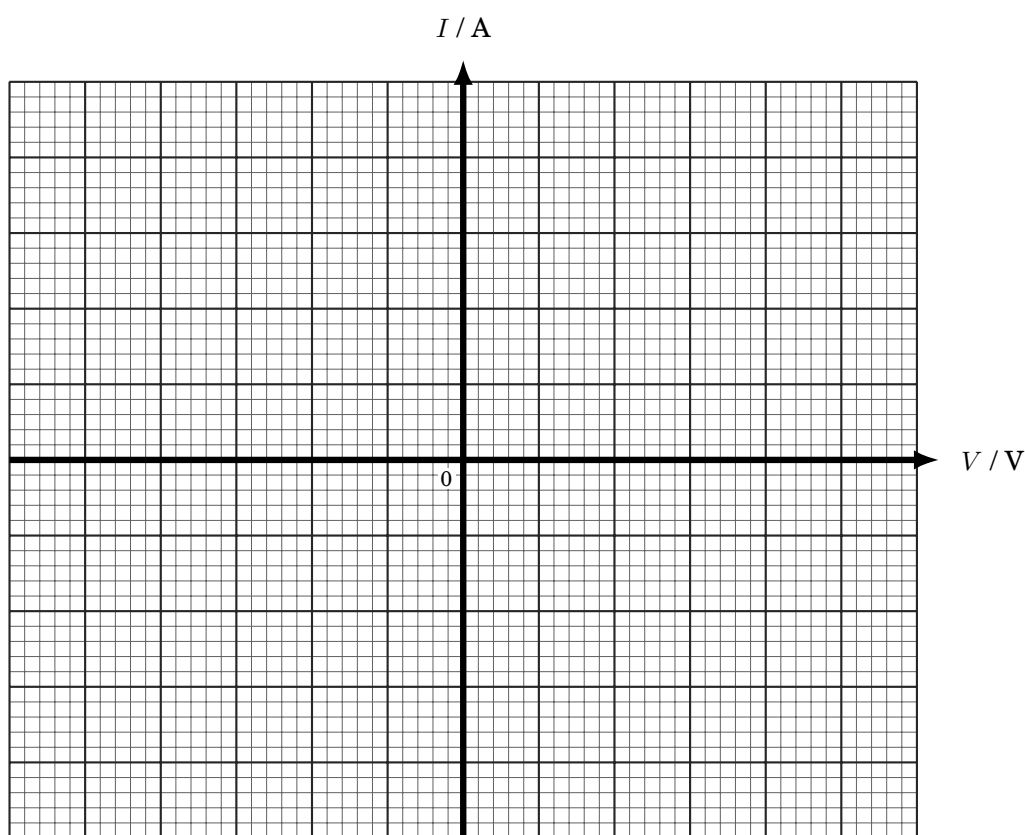
One way round, you will get a current. The other way round, you will get nothing — and nothing is the correct result.

Do not call it a broken circuit, do not go looking for a loose lead, and do not ask for a new diode. A reading of zero with several volts across it is exactly what a diode is *for*. Write the zeros down: they are your data.

Two further things you will notice in the forward direction:

- Almost nothing happens until about **0.6 V**. Below that the diode barely conducts at all.
- Once it passes that point the current climbs very steeply. **Go up in small steps here** or you will get two useful points and a mess.

pd V / V	Current I / A	pd V / V	Current I / A
forward		reversed (negative)	



Label your own scale on both axes first. The two halves of this graph do **not** match — draw what you measured, not what looks tidy.

6) **Describe** what your graph does when the potential difference is reversed.

The current stays at zero. The line lies flat along the horizontal axis for the whole of the negative

side, no matter how large the reversed *pd* becomes.

7) **State** what this tells you about the resistance of a diode in the reverse direction.

It is extremely high — effectively infinite. A large potential difference produces no measurable current at all.

8) **Complete** the sentence. Circle one: **current through a diode flows** _____.

equally in both directions

in one direction only

only above 6 volts

only when it is hot

9) **Suggest** one job a diode could usefully do in a circuit, given what you have just measured.

Any sensible use of one-way conduction: protecting a circuit from a battery inserted the wrong way round; making sure a motor can only turn one way; converting alternating current to direct current by blocking the half of the cycle that runs backwards.

Two more components — no practical needed

Your graphs cover the three components AQA names. Two others change their resistance too, and you need to know what changes it.

Component	What changes its resistance	What it is used for
Thermistor	<i>Temperature. As the temperature rises, its resistance falls.</i>	<i>Thermostats and temperature sensors — ovens, fridges, fire alarms, car engine sensors.</i>
LDR light-dependent resistor	<i>Light intensity. As it gets brighter, its resistance falls.</i>	<i>Switching lights on at dusk — street lamps, security lights, automatic headlights, camera exposure.</i>

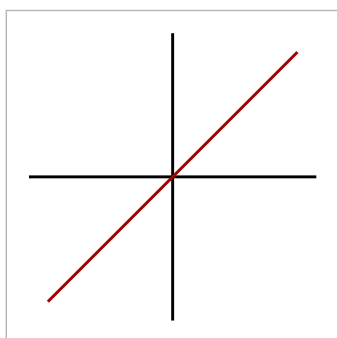
The pattern worth spotting. A fixed resistor has one resistance and keeps it. Every other component on this sheet has a resistance that *depends on something* — the current through it, its temperature, or the light falling on it.

Pulling it together

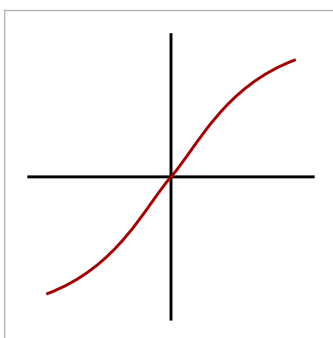
10) **Name** the one component you tested whose resistance stayed constant, and say how the graph showed it.

The fixed resistor. Its graph was a straight line through the origin, so V/I was the same at every point.

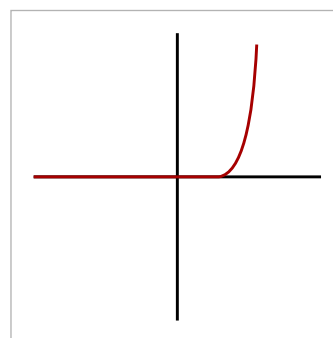
11) **Sketch** all three characteristics side by side, from memory, in the boxes below. Axes only — no numbers.



Fixed resistor



Filament lamp



Diode

Shapes only — no numbers expected. The diode's flat section must sit on the axis to the left of the origin, and the lamp's curve must be symmetrical about the origin.

Evaluation

12) **Explain** why the readings for the fixed resistor would be wrong if you left the switch closed for the whole experiment.

The current would warm the resistor. A warmer resistor has a greater resistance, so the later readings would give a smaller current than they should and the line would bend instead of staying straight — and the component would no longer be “at constant temperature”, which is the condition the experiment is meant to test.

13) **Give** one reason why the filament lamp's graph is a curve while the fixed resistor's is a straight line, even though both are just pieces of metal wire.

The filament gets far hotter. It is a very thin wire designed to glow, so its temperature changes enormously over the range of pd used, and resistance rises with temperature. The fixed resistor barely warms at all, so its resistance stays effectively constant.

14) **Suggest** one improvement to the method that would make the diode's turn-on point easier to identify.

Take many more readings in small steps between about 0.4 V and 0.8 V, rather than using the same large steps as the rest of the range — and repeat them to take a mean. The turn-on happens over a very narrow range of pd, so widely spaced readings step straight over it.

Extra space for working

A second attempt at a graph, a repeated set of readings, anything that did not fit.

Before you pack away: switch off at the supply, disconnect, and check the lamp is *cool* before it goes back in the tray. Diodes back in their own box — they are easy to lose and easy to destroy. Meters back the way you found them.